

BATCH -14

HUMAN FACE DETECTION AND GENDER & AGE PREDICTION
BY AWS AI-ML VIRTUAL INTERNSHIP

Internship-I report submitted in partial fulfilment of requirements for the award of degree of

BACHELOR OF TECHNOLOGY

IN

COMPUTER SCIENCE AND ENGINEERING (AI & ML)

By

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Under the esteemed guidance

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GAYATRI VIDYA PARISHAD COLLEGE OF ENGINEERING (AUTONOMOUS)
MADHURAWADA, VISAKHAPATNAM – 530048
2025 - 2026

**GAYATRI VIDYA PARISHAD COLLEGE OF ENGINEERING (AUTONOMOUS)
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**COLLEGE OF ENGINEERING
(AUTONOMOUS)**

CERTIFICATE

This report on
**“HUMAN FACE DETECTION AND GENDER & AGE PREDICTION
BY AWS AI-ML VIRTUAL INTERNSHIP”**

is a bonafide record of the Internship work submitted by

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In their V semester in partial fulfillment of the requirements for the Award of Degree of

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During the academic year 2025-2026

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अखिल भारतीय तकनीकी शिक्षा परिषद्
All India Council for Technical Education



Certificate of Virtual Internship

This is to certify that

KANNURU KRISHNA VENI

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has successfully completed 10 weeks

AI-ML Virtual Internship

During April - June 2025

We wish him / her all the best for the future endeavours

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GRADE- O (Outstanding):90-100 | E (Excellent):80-89 | A (Very Good):70-79 | B (Good): 60-69 | C (Fair): 50-59 | D (Average): 40-49 | P (Pass): 30-39 | F (Fail): Below 30

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ABSTRACT:

The AWS AI&ML Virtual Internship enabled us to bridge theory with practice through the development of **FaceIQ**, a web-based application for real-time facial analysis. The internship began with **AWS Cloud Foundations**, where we explored the advantages of cloud computing, global infrastructure, pricing models, and security compliance. These fundamentals guided our deployment decisions and cost optimization strategies.

FaceIQ was designed to deliver accessible face recognition without requiring GPUs or paid services, using **pre-trained DeepFace models** that run efficiently on CPUs. Leveraging the **AWS compute module**, we deployed the application on an **Amazon EC2 instance** under the free tier. To ensure global accessibility, we integrated **DuckDNS** to map our public EC2 IP to a free domain. Security was enhanced by configuring **Nginx** via the AWS CLI and enabling **SSL certificates**, ensuring encrypted access through HTTPS.

The project workflow reflected AWS architectural best practices. From **storage and monitoring** using AWS tools to adopting the **shared responsibility model** for securing accounts, each module reinforced deployment robustness. Additional AWS ML foundations, such as **SageMaker pipelines, forecasting, computer vision, NLP, and generative AI**, inspired extensions of our work and validated the feasibility of scalable ML applications.

FaceIQ offers a **user-friendly dashboard** supporting both image uploads and live webcam input. We implemented **confidence thresholds** to filter low-confidence or duplicate detections, ensuring meaningful results. Users can download **annotated images and CSV reports**, enhancing practical usability. Development progressed from **Streamlit prototyping in Anaconda virtual environments**, through **UI testing on desktop and mobile modes**, to final deployment on the cloud.

By applying AWS cloud and ML concepts end-to-end—covering **cost optimization, secure deployment, scalable infrastructure, and ML integration**—FaceIQ demonstrates how academic learning can be transformed into a real-world, publicly available solution

AWS Cloud Foundation

MODULE	Topic name	Pg No.
Module 1: Cloud Concepts Overview	1.1 Introduction to cloud computing	11-12
	1.2 Introduction to Amazon Web Services (AWS)	
Module 2: Cloud Economics and Billing	2.1 Fundamentals of pricing	13
	2.2 AWS Organizations	
	2.3 Technical support	
Module 3: AWS Global Infrastructure Overview	3.1 AWS Global Infrastructure	14
	3.2 AWS services and service category overview	
Module 4: AWS Cloud Security	4.1 AWS shared responsibility model	15-16
	4.2 Securing accounts	
	4.3 Working to ensure compliance	
Module 5: Networking and Content Delivery	5.1 Networking basics	17-18
	5.2 Amazon VPC	
	5.3 Amazon Route 53	
Module 6: Compute	6.1 Compute services overview	19
	6.2 Amazon EC2	
	6.3 Introduction to AWS Elastic Beanstalk	
Module 7: Storage	7.1 Amazon Elastic Block Store (Amazon EBS)	20
	7.2 Amazon Elastic File System (Amazon EFS)	
Module-8: Databases	8.1 Amazon Relational database service	21
	8.2 Amazon DynamoDB	
	8.3 Amazon Redshift	
	8.4 Amazon Aurora	
Module 9: Cloud Architecture	9.1 Introduction to AWS well architected frame designing principles and activity	22
	9.2 Reliability and availability	
Module 10: Automatic Scaling and Monitoring	10.1 Elastic Load Balancing	23
	10.2 Amazon CloudWatch	
	10.3 Amazon EC2 Auto Scaling	

AWS Machine Learning Foundations

MODULE	Topic name	Pg No.
Module 1: Welcome to AWS Academy ML Foundations	1.1 Course Overview and Learning Objectives 1.2 AWS Academy Learning Environment 1.3 Getting Started with AWS Tools	24-25
Module 2: Introducing Machine Learning	2.1 What is Machine Learning? 2.2 Business Problems Solved with ML 2.3 Machine Learning Process and Challenges	26-27
Module 3: ML Pipeline with Amazon SageMaker	3.1 Formulating ML Problems 3.2 Data Collection, Processing & Feature Engineering 3.3 Model Training, Evaluation, and Deployment	28
Module 4: Introducing Forecasting	4.1 Forecasting Concepts and Applications 4.2 Time Series Data Characteristics 4.3 Forecasting using Amazon Forecast	29
Module 5: Introducing Computer Vision	5.1 Introduction to Computer Vision 5.2 Image and Video Analysis Techniques 5.3 Preparing Datasets for Computer Vision	30
Module 6: Introducing Natural Language Processing	6.1 Overview of Natural Language Processing (NLP) 6.2 AWS NLP Managed Services (Comprehend, Polly, Translate) 6.3 Applications of NLP in Real-world Use Cases	31
Module 7: Introducing Generative AI	7.1 What is Generative AI? 7.2 How Generative AI Works (Foundation Models & Prompting) 7.3 AWS Generative AI Services (e.g., Amazon Q, Bedrock)	32

AWS Cloud Foundation

Module 1: Cloud Concepts Overview

1.1 Introduction to cloud computing

Cloud computing enables you to think of your infrastructure as software. Software solutions are flexible. With the cloud computing model, you can treat resources as temporary and disposable.

There are three main cloud service models:

- **Infrastructure as a service (IaaS)**
- **Platform as a service (PaaS)**
- **Software as a service (SaaS)**

There are three main cloud computing deployment models:

- **Cloud**
- **Hybrid**
- **On-premises**

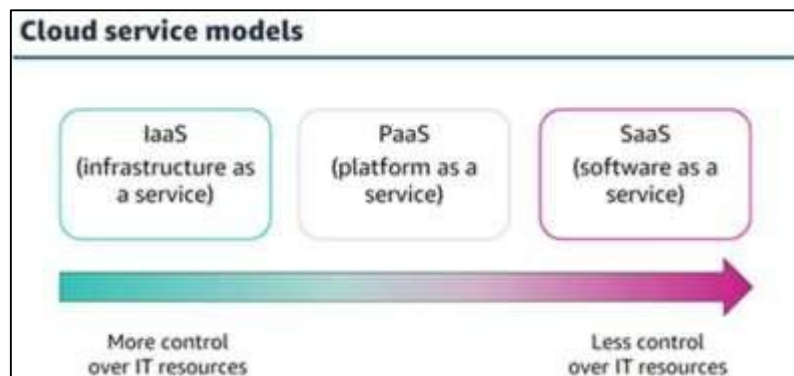


Fig 1.1 Service Models

Advantages of cloud computing

- Trade capital expense for variable expense
- Massive economies of scale
- Stop guessing capacity
- Increase speed and agility
- Stop spending money on running and maintaining data centers
- Go global in minutes

1.2 Introduction to Amazon Web Services (AWS)

- AWS is a secure cloud platform that offers a broad set of global cloud-based product called services that are designed to work together.
- There are many categories of AWS services, and each category has many services to choose from.
- Choose a service based on your business goals and technology requirements.
- There are three ways to interact with AWS services.

AWS Cloud Adoption Framework (AWS CAF)

- Cloud adoption is not instantaneous for most organizations and requires a thoughtful, deliberate strategy and alignment across the whole organization.
- The AWS CAF was created to help organizations develop efficient and effective plans for their cloud adoption journey.
- The AWS CAF organizes guidance into six areas of focus, called perspectives.
- Perspectives consist of sets of business or technology capabilities that are the responsibility of key stakeholders.

Module 2: Cloud Economics and Billing

2.1 Fundamentals of pricing

- Inbound data transfer.
- Data transfer between services within the same AWS Region.
- Pay for what you use.
- Start and stop anytime.

Total Cost of Ownership

- On-premises versus cloud is a question that many businesses ask. The difference between these two options is how they are deployed.
- A cloud infrastructure is purchased from a service provider who builds and maintains the facilities, hardware, and maintenance staff.
- A customer pays for what is used, Scaling-up or down is simple.

2.2 AWS Organizations

- Centrally managed access policies across multiple AWS accounts.
- Controlled access to AWS services.
- Automated AWS account creation and management.
- Consolidated billing across multiple AWS accounts.

AWS Billing and Cost Management

- AWS Billing and Cost Management is the service that you use to pay your AWS bill monitor your usage, and budget your costs.
- Billing and Cost Management enables you to forecast and obtain a better idea of what your costs and usage might be in the future so that you can plan ahead.

2.3 Technical support

- AWS Support was developed to provide complete support and the right resources to aid your success.
- AWS Support has Technical Account Managers (TAMs) who are designated as that user's primary point of contact.

Module 3: AWS Global Infrastructure Overview

3.1 AWS Global Infrastructure

- The AWS Cloud infrastructure is built around Regions.
- AWS has 22 Regions worldwide.
- An AWS Region is a physical geographical location with one or more Availability Zones. Availability Zones in turn consist of one or more data centers.
- To achieve fault tolerance and stability, Regions are isolated from one another.
- Resources in one Region are not automatically replicated to other Regions.
- The AWS Global Infrastructure consists of Regions and Availability Zones.
- Each Availability Zone is physically separate from other Availability Zones and has redundant power, networking, and connectivity.

3.2 AWS services and service category overview

- AWS Global Infrastructure can be broken down into three elements:
- Regions
- Availability Zones
- Points of Presence
- AWS offers a broad set of cloud-based services.

AWS storage services

Amazon Simple Storage Service (Amazon S3), Amazon Elastic Block Store (Amazon EBS), Amazon Simple Storage Service Glacier, Amazon Elastic File System (Amazon EFS).

AWS compute services

Amazon Elastic Compute Cloud (Amazon EC2), Amazon EC2 Auto Scaling, Amazon Elastic Container Service (Amazon ECS), Amazon Elastic Container Registry (Amazon ECR)...etc.

AWS database services

Amazon Relational Database Service (Amazon RDS), Amazon Aurora, Amazon Redshift, Amazon DynamoDB.

AWS networking and content delivery services

Amazon Virtual Private Cloud (Amazon VPC), Elastic Load Balancing, Amazon CloudFront...etc.

AWS security, identity, and compliance services

AWS Identity and Access Management (IAM), AWS Organizations, Amazon Cognito, AWS Artifact, AWS Key Management Service (AWS KMS), AWS Shield.

AWS cost management services

AWS Cost and Usage Report, AWS Budgets, AWS Cost Explorer.

AWS management and governance services

AWS Management Console, AWS Config, Amazon CloudWatch, AWS Auto Scaling,

Module 4: AWS Cloud Security

4.1 AWS shared responsibility model

- AWS is responsible for the physical infrastructure that hosts your resources, including:
- Physical security of data centers
- Hardware infrastructure
- Software infrastructure
- Network infrastructure
- Infrastructure as a service (IaaS) refers to services that provide basic building blocks for cloud IT, typically including access to configure networking, computers (virtual or on dedicated hardware), and data storage space.
- Platform as a service (PaaS) refers to services that remove the need for the customer to manage the underlying infrastructure (hardware, operating systems, etc.).
- Software as a service (SaaS) refers to services that provide centrally hosted software that is typically accessible via a web browser, mobile app, or application programming interface (API).

AWS Identity and Access Management (IAM)

- AWS Identity and Access Management (IAM) allows you to control access to compute, storage, database, and application services in the AWS Cloud.
- IAM can be used to handle authentication, and to specify and enforce authorization policies so that you can specify which users can access which services.
- An IAM user is a person or application that is defined in an AWS account, and that must make API calls to AWS products.
- An IAM group is a collection of IAM users. You can use IAM groups to simplify specifying and managing permissions for multiple users.
- An IAM policy is a document that defines permissions to determine what users can do in the AWS account.
- An IAM role is a tool for granting temporary access to specific AWS resources in an AWS account.

Securing a new AWS account

- Stop using the account root user as soon as possible.
- Enable multi-factor authentication (MFA).
- Use AWS CloudTrail.
- Enable a Billing report, such as the AWS Cost and Usage Report.

4.2 Securing accounts

- Group AWS accounts into organizational units (OUs) and attach different access policies to each OU.
- Integration and support for IAM
- Permissions to a user are the intersection of what is allowed by AWS Organizations and what is granted by IAM in that account.

- Use service control policies to establish control over the AWS services and API actions that each AWS account can access
- Service control policies (SCPs) offer centralized control over accounts.

Securing data on AWS

- Data encryption is an essential tool to use when your objective is to protect digital data.
- Data encryption takes data that is legible and encodes it so that it is unreadable to anyone who does not have access to the secret key that can be used to decode it.
- Data in transit refers to data that is moving across the network.
- AWS Certificate Manager is a service that enables you to provision, manage, and deploy SSL or TLS certificates for use with AWS services and your internal connected resources.

4.3 Working to ensure compliance

- AWS engages with external certifying bodies and independent auditors to provide customers with information about the policies, processes, and controls that are established and operated by AWS.
- AWS security compliance programs provide information about the policies, processes, and controls that are established and operated by AWS.
- AWS Config is used to assess, audit, and evaluate the configurations of AWS resources.

Module 5: Networking and Content Delivery

5.1 Networking basics

- A network can be logically partitioned into subnets.
- Networking requires a networking device (such as a router or switch) to connect all the clients together and enable communication between them.
- Each client machine in a network has a unique Internet Protocol (IP) address that identifies it. An IP address is a numerical label in decimal format.
- Machines convert that decimal number to a binary format. In this example, the IP address is 192.0.2.0.

5.2 Amazon VPC

- An elastic network interface is a virtual network interface that you can attach or detach from an instance in a VPC.
- A network interface's attributes follow it when it is reattached to another instance.
- When you move a network interface from one instance to another, network traffic is redirected to the new instance.
- A route table contains a set of rules (called routes) that directs network traffic from your subnet. Each route specifies a destination and a target.

VPC networking

- An internet gateway is a scalable, redundant, and highly available VPC component that allows communication between instances in your VPC and the internet.
- An internet gateway serves two purposes: to provide a target in your VPC route tables for internet-routable traffic.
- A network address translation (NAT) gateway enables instances in a private subnet to connect to the internet or other AWS services, but prevents the internet from initiating a connection with those instances.

VPC security

- A network access control list (network ACL) is an optional layer of security for your Amazon VPC.
- It acts as a firewall for controlling traffic in and out of one or more subnets.
- A network ACL contains a numbered list of rules that are evaluated in order, starting with the lowest numbered rule.
- Security groups act at the instance level, but network ACLs act at the subnet level.
- Security groups support allow rules only, but network ACLs support both allow and deny rules.

5.3 Amazon Route 53

- Amazon Route 53 is a highly available and scalable cloud Domain Name System (DNS) web service.
- Amazon Route 53 effectively connects user requests to infrastructure running in AWS such as Amazon EC2 instances, Elastic Load Balancing load balancers, or Amazon S3 buckets and can also be used to route users to infrastructure that is outside of AWS.

Amazon CloudFront

- A content delivery network(CDN) is a globally distributed system of caching servers.
- A CDN caches copies of commonly requested files

Open Systems Interconnection (OSI) model			
Layer	Number	Function	Protocol/Addresses
Application	7	Means for an application to access a computer network	HTTP(S), FTP, DHCP, LDAP
Presentation	6	• Ensures that the application layer can read the data • Encryption	ASCII, ICA
Session	5	Enables orderly exchange of data	NetBIOS, RPC
Transport	4	Provides protocols to support host-to-host communication	TCP, UDP
Network	3	Routing and packet forwarding (routers)	IP
Data link	2	Transfer data in the same LAN network (hubs and switches)	MAC
Physical	1	Transmission and reception of raw bitstreams over a physical medium	Signals (1s and 0s)

Fig 5.1 OSI Model

Module 6: Compute

6.1 Compute services overview

- Amazon EC2 provides virtual machines, and flexibility and leave many of the server management responsibilities to you.
- AWS Lambda is a zero administration compute platform. It enables you to run code without provisioning or managing servers.
- Container based services including Amazon Elastic Kubernetes Service, and Amazon Elastic Container Registry enable you to run multiple workloads on a single operating system (OS).

6.2 Amazon EC2

- Elastic refers to the fact that you can easily increase or decrease the number of servers you run to support an application automatically.
- Compute refers to host running applications or process data actions that require compute resources, including processing power (CPU) and memory (RAM).
- An Amazon Machine Image (AMI) provides information that is required to launch an EC2 instance.
- Amazon Elastic File System (Amazon EFS) provides a simple, scalable, fully managed elastic Network File System (NFS) file system for use with AWS Cloud services and on premises resources.

✦ Amazon EC2 cost optimization

- On Demand Instances offer the most flexibility, with no long term contract and low rates.
- Spot Instances provide large scale at a significantly discounted price.
- Reserved Instances are a good choice if you have predictable compute needs.
- Dedicated Hosts are a good choice when you have licensing restrictions for the software you want to run on Amazon EC2.

To optimize costs, you must consider four consistent, powerful drivers:

- Rightsize
- Increase elasticity
- Optimal pricing model
- Optimize storage choices

Container services:

- Docker is a software platform that packages software (such as applications) into containers.
- The right of the diagram has a virtual machine (VM)based deployment. Each of the three EC2 instances runs directly on the hypervisor.
- The left of the diagram has a container based deployment. There is only one EC2 instance that runs a virtual machine.
- Amazon Elastic Kubernetes Service (Amazon EKS) is a managed Kubernetes service that makes it easy for you to run Kubernetes on AWS

6.3 Introduction to AWS Elastic Beanstalk:

- AWS Elastic Beanstalk is another AWS compute service option and a platform that facilitates the quick deployment and management of your web applications and services.
- Elastic Beanstalk enables you to retain full control over the AWS resources that power your application. Elastic Beanstalk is fast and simple to start using.
- Elastic Beanstalk automatically handles the deployment details of capacity provisioning, load balancing, automatic scaling, and monitoring application health.

Module 7: Storage:

7.1 Amazon Elastic Block Store (Amazon EBS)

- Amazon EBS provides persistent block storage volumes for use with Amazon EC2 instances. It is also sometimes called non-volatile storage.
- Block storage solutions are typically faster and use less bandwidth, but they can cost more than Object-level storage.

Amazon Elastic File System (Amazon EFS)

- Amazon Elastic File System (Amazon EFS) provides simple, scalable, elastic file storage for use with AWS services and on premises resources.
- It can automatically scale from gigabytes to petabytes of data without the need to provision storage.
- With Amazon EFS, you can create the following on an Amazon EC2 instance:
 - ID
 - Creation token
 - Creation time
 - File system size in bytes

7.2 Amazon S3 Glacier

By using lifecycle policies, you can cycle data at regular intervals between different Amazon S3 storage types. This automation reduces your overall cost, because you pay less for data as it becomes less important with time. In addition to setting lifecycle rules per object, you can also set lifecycle rules.

Storage comparison		
	Amazon S3	Amazon S3 Glacier
Data Volume	No limit	No limit
Average Latency	ms	minutes/hours
Item Size	5 TB maximum	40 TB maximum
Cost/GB per Month	Higher cost	Lower cost
Billed Requests	PUT, COPY, POST, LIST, and GET	UPLOAD and retrieval
Retrieval Pricing	\$ Per request	\$\$ Per request and per GB

Fig 7.1 Storage comparisons

Module-8: Databases:

8.1 Amazon Relational database service:

Amazon RDS is a managed service that sets up and operates a relational database in the cloud. Amazon RDS enables you to focus on your application, so you can give applications the performance, high availability, security, and compatibility that they need.

Amazon RDS DB instances - DB Instance Classes:

- CPU
- Memory
- Network performance

DB Instance Storages:

- Magnetic
- General Purpose (solid state drive, or SSD)
- Provisioned IOPS

Amazon RDS read replicas

8.2 Amazon DynamoDB:

DynamoDB is a fast and flexible NoSQL database service for all applications that need consistent, single-digit-millisecond latency at any scale

- The core DynamoDB components are tables, items, and attributes.
- DynamoDB supports two different kinds of primary keys. The partition key is a simple primary key, which is composed of one attribute called the sort key.
- NoSQL database tables
- Virtually unlimited storage

8.3 Amazon Redshift

- Amazon Redshift is a fast, fully managed data warehouse that makes it simple and cost-effective to analyze all your data by using standard SQL and your existing business intelligence (BI) tools.
- Amazon Redshift is designed to consistently deliver high performance. Amazon Redshift uses columnar storage and a massively parallel processing architecture. These features parallelize and distribute data and queries across multiple nodes.
- Amazon Redshift also automatically monitors your cluster and backs up your data so that you can easily restore if needed. Encryption is built in—you only need to enable it.

8.4 Amazon Aurora

- Amazon Aurora is a highly available, performant, and cost-effective managed relational database.
- Aurora offers a distributed, high-performance storage subsystem. Using Amazon Aurora can reduce your database costs while improving the reliability of the database.
- Multiple levels of security are available, including network isolation by using Amazon VPC encryption at rest by using keys that you create and control through AWS Key Management Service (AWS KMS); and encryption of data in transit by using Secure Sockets Layer (SSL)
- This reduces the need for you to throttle access until the cache is repopulated to avoid brownouts.

Module 9: Cloud Architecture:

Architecture is the art and science of designing and building large structures. Large systems require architects to manage their size and complexity.

Introduction to AWS well architected frame designing principles and activity

- The AWS Well-Architected Framework is a guide that is designed to help you build the most secure, high-performing, resilient, and efficient infrastructure possible for your cloud applications and workloads
- AnyCompany Corporation sells high-quality three-dimensional (3D) printed cityscapes of neighborhoods that enable you to see individual buildings and trees.
- AWS Well-Architected Framework design principles for each of the following five pillars: operational excellence, security, reliability, performance efficiency, and cost optimization.
- AnyCompany Corporation has three main departments:
- Fly and Snap– image acquisition, preprocessing, and storage
- Show and Sell– promoting, selling, and working with customers
- Make and Ship– manufacturing of products and delivery

Operational excellence design principles

- Perform operations as code
- Make frequent, small, reversible changes
- Refine operations procedures frequently

9.2 Reliability and availability:

- A measure of your system's ability to provide functionality when desired by the user.
- Probability that your entire system will function as intended for a specified period.
- Formally, Availability is the percentage of time that a system is operating normally or correctly performing the operations expected of it
- A highly available system is one that can withstand some measure of degradation while still remaining available. In a highly available system, downtime is minimized as much as possible and minimal human intervention is required.

Module 10: Automatic Scaling and Monitoring

10.1 Elastic Load Balancing

Elastic Load Balancing is an AWS service that distributes incoming application or network traffic across multiple targets—such as Amazon Elastic Compute Cloud (Amazon EC2) instances, containers, internet protocol (IP) addresses, and Lambda functions—in a single Availability Zone or across multiple Availability Zones.

- Elastic Load Balancing distributes incoming application or network traffic across multiple in one or more Availability Zones.
- Elastic Load Balancing supports three types of load balancers:
 - Application Load Balancer
 - Network Load Balancer

10.2 Amazon CloudWatch

- Amazon CloudWatch helps you monitor your AWS resources—and the applications that you run on AWS—in real time.
- CloudWatch enables you to -.
- Set alarms to automatically send notifications to SNS topics or perform Amazon EC2 Auto Scaling.
- Define rules that match changes in your AWS environment and route these events to targets for processing.

10.3 Amazon EC2 Auto Scaling

- Scaling enables you to respond quickly to changes in resource needs.
- Amazon EC2 Auto Scaling helps you maintain application availability, and enables you to automatically add or remove EC2 instances according to your workloads.
- An Auto Scaling group is a collection of EC2 instances. AWS Auto Scaling is a separate service that monitors your applications, and it automatically adjusts capacity.

AWS Academy Machine Learning Foundations

Module 1: Welcome to AWS Academy ML Foundations

1.1 Course Overview and Learning Objectives

This course serves as a foundational introduction to machine learning (ML) and its practical implementation using Amazon Web Services (AWS). The primary goal is to equip learners with both theoretical and hands-on skills in developing and deploying ML models. By the end of the course, learners should be able to:

- Understand the fundamental principles of ML, including supervised, unsupervised, and reinforcement learning.
- Recognize appropriate ML algorithms for specific tasks.
- Prepare and preprocess datasets effectively.
- Train, evaluate, and optimize ML models using AWS tools.
- Deploy models to cloud environments using Amazon SageMaker.
- The course is structured around real-world use cases and integrates AWS-managed services to simulate industry-standard practices.

1.2 AWS Academy Learning Environment

The AWS Academy Learning Environment is a comprehensive educational platform that provides students with access to learning materials and hands-on lab experiences using AWS services. The platform is designed to mirror real-world cloud environments, allowing learners to:

- Access and manage cloud-based lab instances.
- Watch instructional videos and demonstrations.
- Complete interactive quizzes and assignments.
- Gain practical experience through guided labs.
- This environment supports experiential learning and prepares students to confidently work in professional cloud computing and ML settings.

1.3 Getting Started with AWS Tools

Learners are introduced to key AWS tools necessary for developing machine learning models in the cloud. These include:

- Amazon S3 (Simple Storage Service): Used for storing large datasets that ML models require.
- Amazon SageMaker: A fully managed service that provides tools for building, training, and deploying ML models quickly.

- AWS IAM (Identity and Access Management): Used to securely manage user access and permissions.
- AWS Management Console: A web-based interface to access and control AWS services.
- The initial setup includes configuring credentials, launching lab environments, and navigating between services. Understanding the core AWS ecosystem is critical before working on ML tasks in this course.

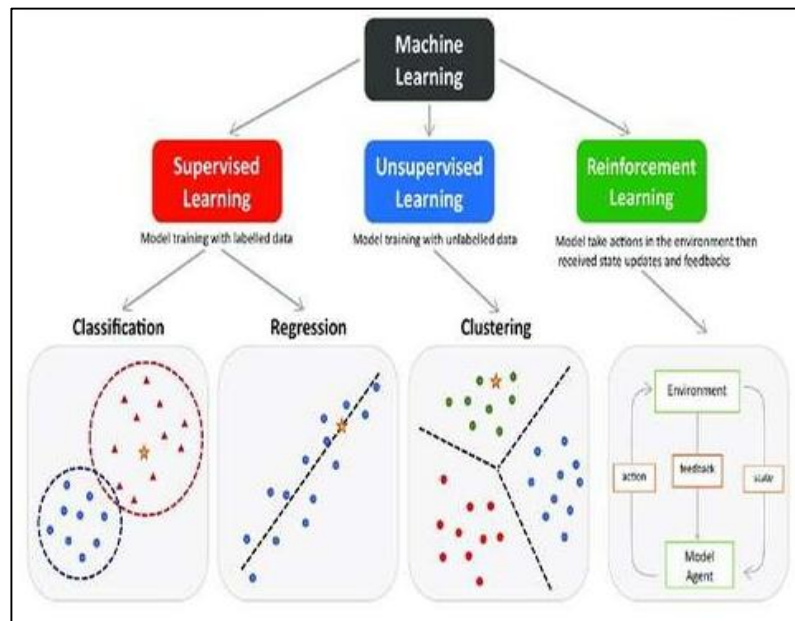


Fig:1.1 Types of Machine Learning

Module 2: Introducing Machine Learning

2.1 What is Machine Learning?

Machine Learning (ML) is a core subfield of Artificial Intelligence (AI) that enables computers to learn from data and make predictions or decisions without being explicitly programmed. Instead of writing code for every possible situation, ML models learn patterns and relationships from historical data. This empowers systems to improve their performance over time as they are exposed to more data. ML is broadly classified into three types: supervised learning, unsupervised learning, and reinforcement learning. These categories differ based on how the data is structured and the nature of the learning task.

Key Points:

- ML allows computers to learn from data and improve performance automatically.
- It includes supervised, unsupervised, and reinforcement learning.
- It is widely used in various industries including healthcare, finance, and retail.

2.2 Business Problems Solved with ML

Organizations across the globe use ML to solve a range of complex business problems. By analyzing large volumes of data, ML models can automate decision-making, enhance productivity, and offer personalized services. For example, e-commerce platforms use ML for recommendation systems, banks use it for fraud detection, and telecom companies use it for churn prediction. These applications not only reduce human effort but also enable companies to scale their services efficiently and cost-effectively.

Key Points:

- ML helps solve real-world business problems such as fraud detection and customer churn.
- It automates decisions and increases efficiency in operations.
- It offers personalized experiences based on user behavior.

2.3 ML Process and Challenges

Developing an ML solution involves a systematic process that includes problem formulation, data collection, data preprocessing, model selection, training, evaluation, and deployment. While this process is powerful, it comes with challenges. These include dealing with poor-quality data, selecting the right algorithm, avoiding model overfitting or underfitting, and ensuring ethical and unbiased outcomes. A solid understanding of this pipeline is crucial to successfully applying ML in real-world scenarios.

Key Points:

- The ML workflow includes data preparation, training, evaluation, and deployment.
- Common challenges include bias, poor data quality, and overfitting.
- Proper planning and monitoring are required for reliable results.

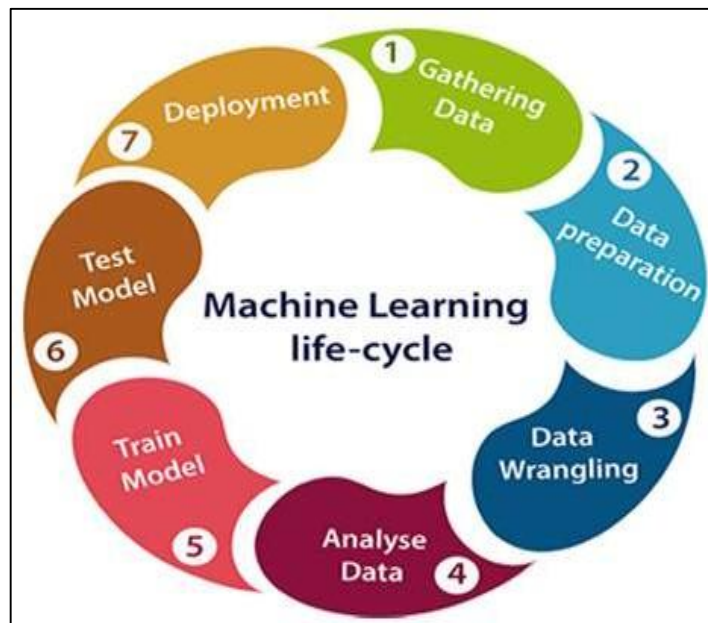


Fig:2.1 ML Life cycle

Module 3: Implementing an ML Pipeline with Amazon SageMaker

3.1 Formulating ML Problems

The first step in any ML project is to correctly frame the problem. This involves identifying what needs to be predicted, selecting the correct type of ML task (e.g., classification, regression), and defining measurable success criteria. A clear understanding of the problem leads to better data collection, feature selection, and model choice.

Key Points:

- Framing the ML problem is critical to the success of the solution.
- It involves selecting the target variable, input features, and model type.
- Understanding business goals helps define the problem accurately.

3.2 Data Collection, Processing & Feature Engineering

Data is the fuel for any ML model. High-quality, relevant data must be collected, cleaned, and processed before model training. Feature engineering involves selecting and transforming variables to make them more useful to the algorithm. AWS services like Amazon S3 and AWS Glue help manage and process data at scale.

Key Points:

- Data preprocessing includes handling missing values, encoding, and normalization.
- Feature engineering improves model performance and accuracy.
- AWS Glue and SageMaker simplify these tasks for cloud-based projects.

3.3 Model Training, Evaluation, and Deployment

Once data is ready, the model is trained using suitable algorithms. Evaluation is performed using metrics like accuracy, precision, recall, and F1 score. A well-performing model is then deployed using services like Amazon SageMaker Endpoints. Deployment enables real-time or batch predictions in production environments.

Key Points:

- Model training involves using data to learn predictive patterns.
- Evaluation helps ensure the model performs reliably.
- SageMaker supports easy deployment and scaling of ML models.

Module 4: Introducing Forecasting

4.1 Forecasting Concepts and Applications

Forecasting is the technique of using past data to make predictions about the future. It is crucial in business planning for sales, inventory management, and financial forecasting. Forecasting models aim to identify patterns such as trends and seasonal effects in time series data.

Key Points:

- Forecasting uses historical data to predict future outcomes.
- It is vital for planning and resource optimization.
- Common applications include sales forecasting and inventory management.

4.2 Time Series Data Characteristics

Time series data is structured by time and often includes patterns like trends (long-term increase/decrease), seasonality (regular fluctuations), and noise (random variation). Understanding these elements is essential to select the correct forecasting model and preprocessing techniques.

Key Points:

- Time series data has order and is dependent on time intervals.
- Trend, seasonality, and noise affect data predictability.
- Analyzing these patterns improves forecast accuracy.

4.3 Forecasting using Amazon Forecast

Amazon Forecast is a fully managed service that automates the ML forecasting process. It uses historical time series data to train models and deliver highly accurate forecasts. The service supports importing data from Amazon S3, configuring forecasting domains, and visualizing predictions.

Module 5: Introducing Computer Vision

5.1 Introduction to Computer Vision

Computer Vision (CV) is the field of ML that deals with analyzing and interpreting visual data like images and videos. It enables machines to recognize objects, classify images, and even understand scenes. CV has revolutionized industries like healthcare, automotive, and retail by automating visual tasks.

Key Points:

- CV allows computers to understand and interpret visual data.
- Tasks include object detection, facial recognition, and image classification.
- It's widely used in healthcare, surveillance, and autonomous vehicles.

5.2 Image and Video Analysis Techniques

Techniques like image segmentation, object detection, and motion tracking are used to extract insights from visual media. AWS offers Amazon Rekognition for pre-trained image and video analysis tasks, reducing the need for complex model training.

Key Points:

- Amazon Rekognition offers scalable, real-time analysis of visual data.
- Image processing can be used for identity verification, surveillance, etc.
- Pre-built APIs make implementation faster and more efficient.

5.3 Preparing Datasets for Computer Vision

Effective model training in CV requires large, annotated datasets. This involves collecting diverse image samples, labeling them correctly, and applying transformations (like rotation or zoom) to improve model generalization. AWS Ground Truth can automate dataset labeling using human input and ML.

Module 6: Introducing Natural Language Processing (NLP)

6.1 Overview of NLP

Natural Language Processing (NLP) allows computers to interact with and derive meaning from human language. It is used for understanding, analyzing, and generating textual or spoken language. NLP is foundational for developing chatbots, virtual assistants, and automated translation systems.

Key Points:

- NLP helps machines process human language.
- Applications include speech recognition, sentiment analysis, and translation.
- It bridges the gap between structured computer logic and human communication.

6.2 AWS NLP Services: Comprehend, Polly, Translate

AWS provides several services for NLP tasks:

- **Amazon Comprehend:** Extracts sentiments, topics, and entities from text.
- **Amazon Polly:** Converts text into realistic speech for voice interfaces.
- **Amazon Translate:** Performs real-time language translation.

Key Points:

- These services simplify NLP deployment without custom model training.
- They support multilingual systems, accessibility, and automation.
- Use cases range from call centers to content generation.

6.3 Applications of NLP in Real-world Use Cases

NLP enables businesses to automate customer support, summarize documents, detect spam, and analyze user feedback. It reduces manual work and enhances user experience through smarter communication tools.

Key Points:

- NLP improves automation in customer service and content management.
- It provides valuable insights from unstructured text data.
- Used in chatbots, document search, and email filtering.

Module 7: Introducing Generative AI

7.1 What is Generative AI?

Generative AI refers to a class of ML techniques that can create new content, such as text, images, audio, or code. Unlike traditional AI that classifies or predicts, generative AI produces original outputs by learning the underlying structure of the input data.

Key Points:

- Generative AI creates new content based on training data.
- It is used in chatbots, image generation, and creative design tools.
- It relies on deep learning and large language models (LLMs).

7.2 How Generative AI Works (Foundation Models & Prompting)

Generative AI uses foundation models trained on massive datasets. These models can perform a wide range of tasks with minimal supervision. Prompt engineering involves crafting inputs to control and fine-tune the model's outputs effectively.

Key Points:

- Foundation models like GPT and Claude are versatile and powerful.
- Prompting controls what the model generates.
- They support many applications across domains using one model.

7.3 AWS Generative AI Services (e.g., Amazon Q, Bedrock)

1. **Amazon Bedrock** provides access to top generative models via APIs, making it easy to embed generative AI into applications.
2. **Amazon Q Developer** assists in software development through code generation and optimization.

Key Points:

- AWS enables secure, scalable access to generative models.
- Services like Bedrock and Q simplify integration into enterprise apps.
- They accelerate productivity and innovation in AI development.

CASE STUDY

The Problem Statement:

With the increasing use of digital services, there is a growing demand for fast, reliable, and accessible face analysis tools. Most existing solutions are prohibitively expensive, requiring powerful hardware (like GPUs) and complex setups, which makes them impractical for small teams, students, or organizations with limited resources. Manual face analysis is slow, subjective, and prone to errors. The problem is to create a cost-effective, easily accessible, and user-friendly web-based application for real-time face analysis.

Finding a Solution

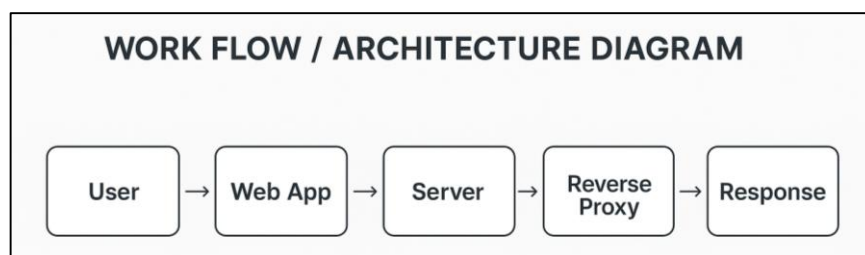
To address these challenges, we developed “FaceIQ - a web-based app” that brings real-time face analysis to everyone, without the need for expensive hardware or paid services.

WORK FLOW / ARCHITECTURE DIAGRAM

The following diagram illustrates the workflow and architecture of the FaceIQ application.

Explanation:

1. **User Interaction:** A user interacts with the FaceIQ web app from any device (desktop or mobile) via their browser. They either upload an image or use their webcam.
2. **Request to Server:** The request is sent to the application server hosted on an **AWS EC2** instance.
3. **Reverse Proxy (Nginx):** The traffic first hits **Nginx**, which acts as a reverse proxy. It listens on port 80 (HTTP) and 443 (HTTPS) and forwards the traffic to the Streamlit app running on port 8501. **Certbot** is used to manage the SSL certificate, enabling a secure HTTPS connection.
4. **Application Processing (Streamlit & DeepFace):** The Python application, built with **Streamlit**, receives the request. It uses the **DeepFace** library to perform face detection, alignment, and analysis (age, gender, emotion). This processing is performed on the CPU, eliminating the need for a GPU.
5. **Data Processing & Refinement:** The application's custom logic applies confidence thresholds and removes duplicate detections to ensure accurate and reliable results.
6. **Response & Output:** The processed data is sent back to the user's browser. The results are displayed on the dashboard, and the user has options to download annotated images and CSV reports.
7. **Domain Management (DuckDNS):** A free **DuckDNS** domain maps the user-friendly URL (e.g., `faceiq.duckdns.org`) to the dynamic public IP address of the AWS EC2 instance.



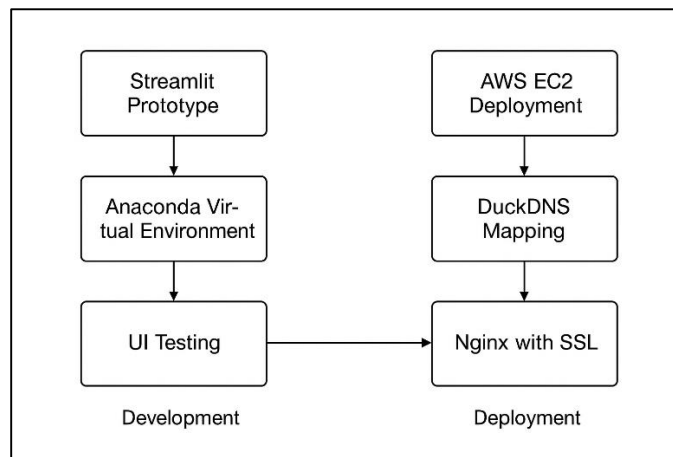


Fig: Life cycle/Phase Cycle of FaceIQ

TECHNIQUES USED:

- **Face Analysis Library:** **DeepFace** is the core library used for face detection, age prediction, gender recognition, and emotion analysis.
- **Web Framework:** **Streamlit** is used to build the interactive and responsive web dashboard with a simple, Python-based interface.
- **Cloud Platform:** **AWS EC2** is the cloud computing service that hosts the application on a free-tier Ubuntu instance.
- **Domain & DNS:** **DuckDNS** provides a free dynamic DNS service, mapping a custom subdomain to the server's public IP.
- **Security:** **SSL/TLS** encryption is implemented using **Certbot** and **Let's Encrypt** to secure data transmission. **Nginx** serves as a reverse proxy to manage secure connections.

METHODOLOGY

The development followed an agile, phased approach:

1. **Local Prototyping:** The application was initially built and tested in a local **Anaconda virtual environment** using Python and Streamlit to ensure all dependencies and features worked correctly.
2. **Code Versioning:** All project files were managed and pushed to a **GitHub** repository, enabling easy deployment and version control.
3. **Cloud Deployment:** An **AWS EC2 (Free Tier)** instance was launched to host the application, demonstrating a cost-effective solution.
4. **Domain Configuration:** A free domain was obtained from **DuckDNS** and linked to the EC2 instance's IP.
5. **Security Implementation:** **Nginx** was set up as a reverse proxy, and **Certbot** was used to automate the process of obtaining and renewing SSL certificates, ensuring secure data transfer via HTTPS.
6. **Final Verification:** The entire system was tested on various devices to confirm full functionality, responsiveness, and security.

Development of FaceIQ:

Phase 1. Development & Local Testing:

a. Building the App in Streamlit

- Developed the FaceIQ dashboard using Python and Streamlit for rapid prototyping and interactive UI.
- Integrated DeepFace for face detection, age, gender, and emotion (smile) prediction.
- Added custom logic for confidence thresholds, duplicate removal, and user-friendly metrics.

b. Running Locally in Anaconda Virtual Environment

- Created a new Anaconda environment for dependency management:

Bash (CLI)

```
conda create -n faceiq python=3.10
```

```
conda activate faceiq
```

```
pip install -r requirements.txt
```

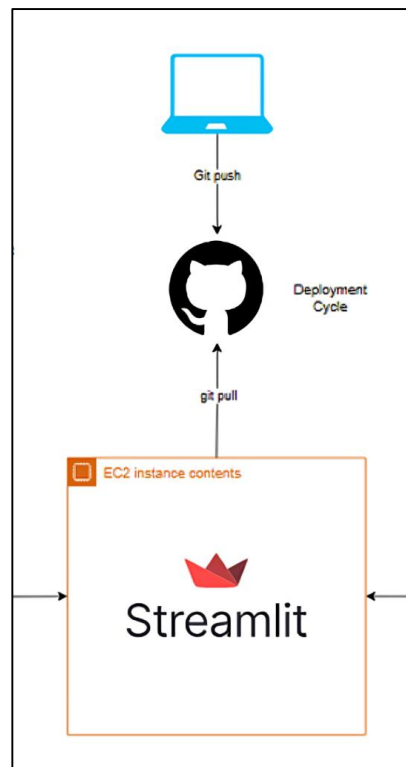
- Launched the app locally:

Bash (CLI):

```
streamlit run app.py
```

c. Device Compatibility & UI Testing

- Tested the app on both desktop and mobile browsers.
- Improved the UI for responsiveness, dark/light theme adaptation, and mobile-friendly layouts.
- Ensured webcam and file upload work smoothly on all devices.
- Upload all files to Github for further process and deployment.



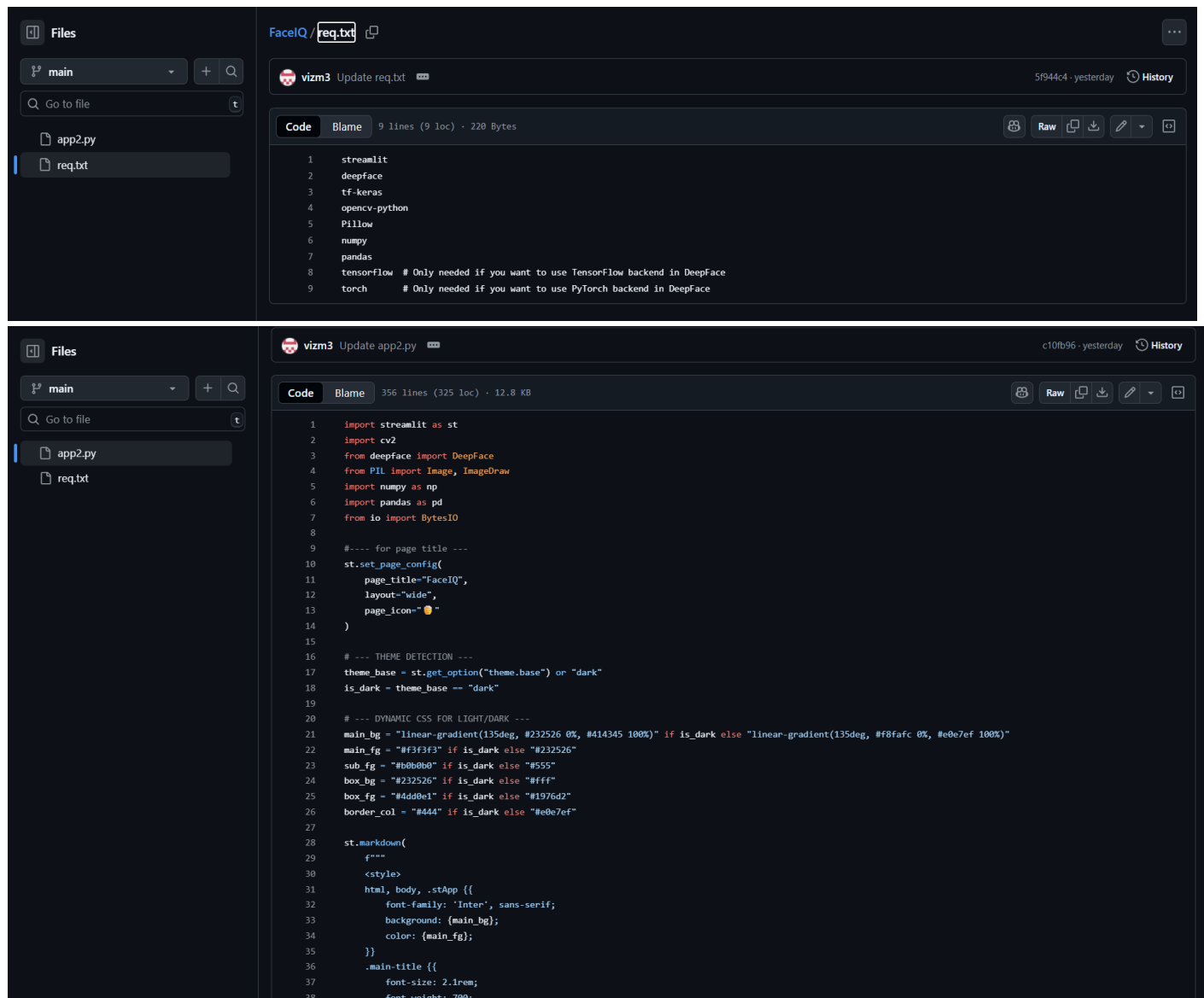


Fig 1.1 Files pushed in Github Account for deployment

Phase 2. Cloud Deployment & Domain Setup:

a. Launching on AWS EC2 (Free Tier)

- Created a free-tier Ubuntu EC2 instance on AWS.
- Opened necessary ports (8501 for Streamlit, 80/443 for web access) in AWS security groups.
- Cloned project from **GITHUB**.

Bash (CLI IN SERVER):

sudo apt update

sudo apt install git

git clone https://github.com/vizm3/FaceIQ.git

cd faceiq

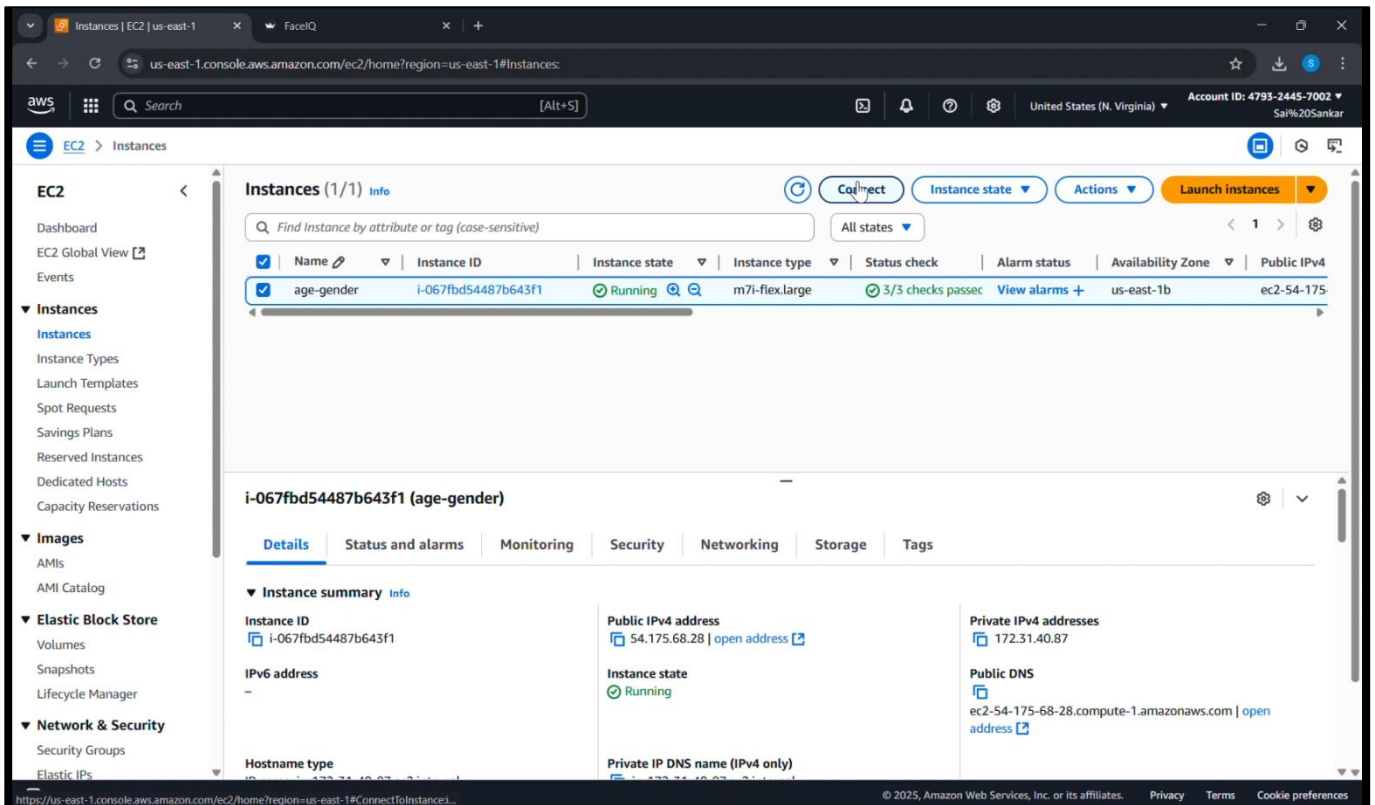


Fig 2.1 Launching an EC2 INSTANCE

b. Mapping a Free Domain with DuckDNS

- Registered a free subdomain at [DuckDNS](https://duckdns.org) (e.g., faceiq.duckdns.org).
- Added the EC2 public IP to the DuckDNS dashboard for the chosen subdomain.
- Set up a cron job or DuckDNS client on the server to keep the IP updated (optional but recommended for dynamic IPs).

DuckDNS Setup Steps:

1. Go to duckdns.org, sign in with GitHub/Google.
2. Choose a subdomain (e.g., faceiq) and add your EC2 public IP.
3. Update your domain's A record if needed (DuckDNS handles this automatically).
4. Test: Visit <http://faceiq.duckdns.org:8501> to access your app.

Phase 3. Enabling HTTPS with SSL (Let's Encrypt & Nginx):

a. Installing Nginx and Certbot

- Installed Nginx as a reverse proxy to forward traffic from ports 80/443 to 8501.
- Installed Certbot for free SSL certificates:

Bash (CLI):

```
sudo apt update
```

```
sudo apt install nginx certbot python3-certbot-nginx
```

b. Configuring Nginx

- Set up an Nginx config to proxy faceiq.duckdns.org to the Streamlit app:

```
server {  
  listen 80;  
  server_name faceiq.duckdns.org;  
  
  location / {  
    proxy_pass http://localhost:8501;  
    proxy_set_header Host $host;  
    proxy_set_header X-Real-IP $remote_addr;  
    proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;  
    proxy_set_header X-Forwarded-Proto $scheme;  
  }  
}
```

c. Obtaining and Enabling SSL

- Ran Certbot to get and install the SSL certificate:

Bash:

```
sudo certbot --nginx -d faceiq.duckdns.org
```

Now, the app is accessible securely at <https://faceiq.duckdns.org>.

Phase 4. Final Deployment Checks & Usage:

a. Final Checks

- Verified the app works on both *desktop and mobile* at <https://faceiq.duckdns.org>.
- Confirmed that the camera input is allowed (since HTTPS is enabled).
- Ensured all features (upload, webcam, download, metrics) work as expected.

b. Demo & Usage

- Users can visit the URL, upload or capture images, and instantly see face analysis results.
- Downloadable annotated images and CSV reports are available for further use.

The UI and responses achieved:

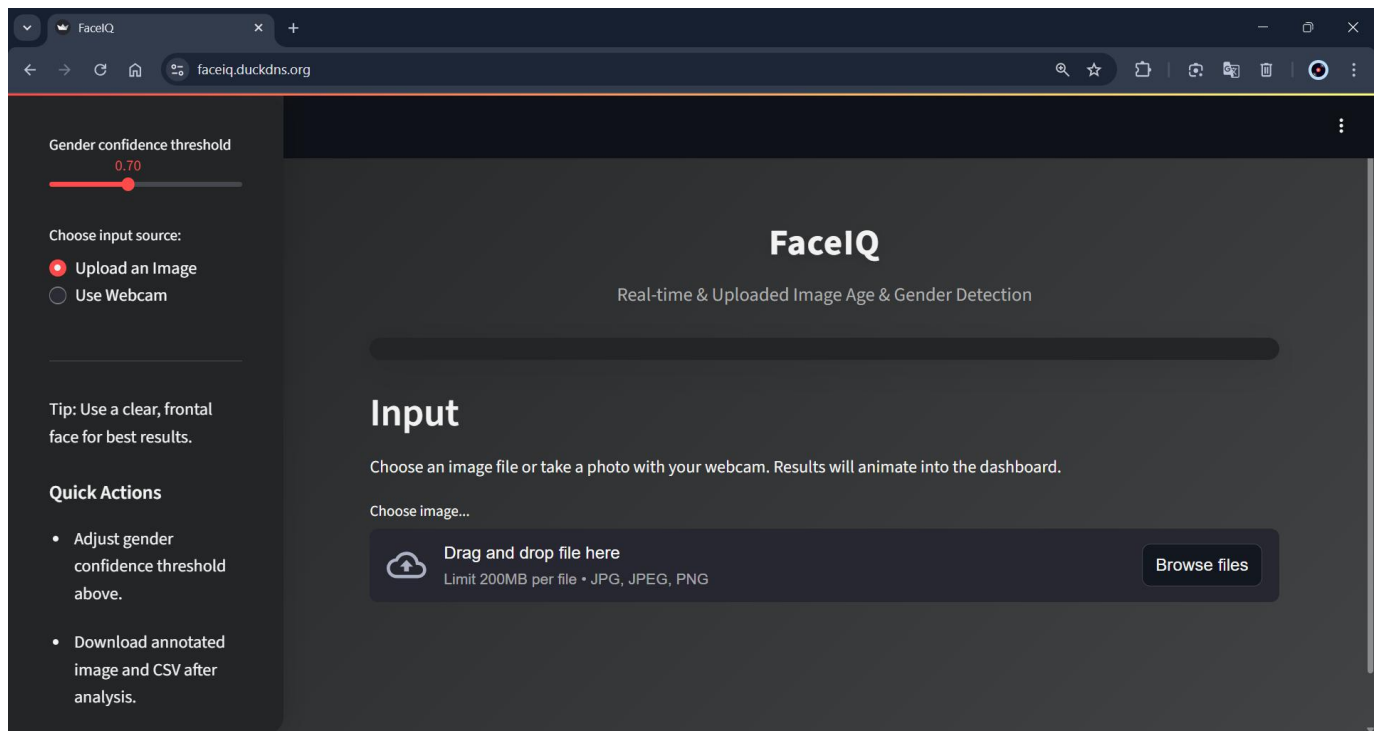


Fig 4.1 The Landing UI

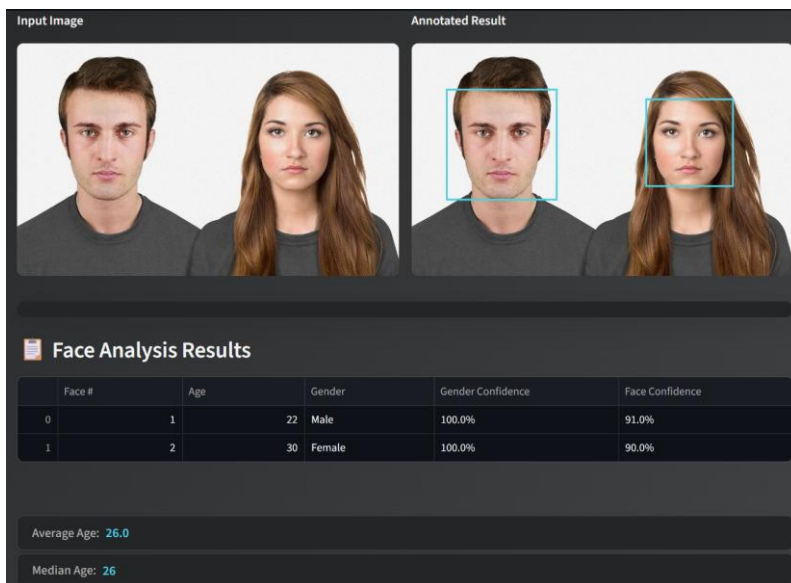


Fig 4.2: The Faces recognized and Classified

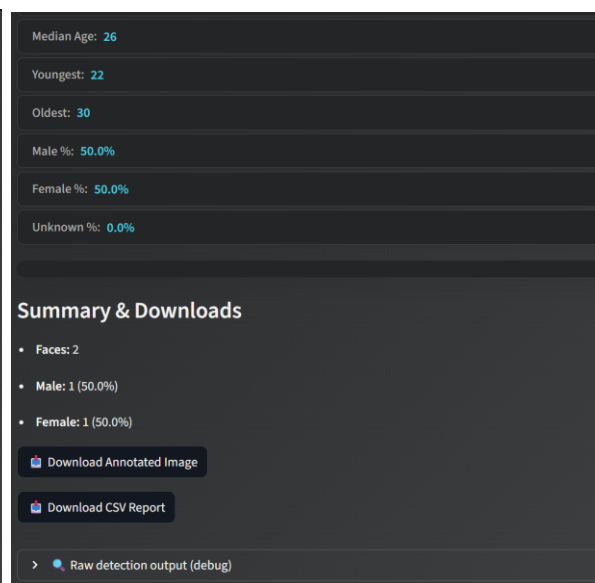


Fig 4.3: Summarization of picture

Use Cases

- **Academic and Workplace Attendance** – Automating attendance with minimal manual intervention.
- **Authentication Gateways** – Serving as a secondary layer of user verification for secure systems.
- **Photo Archiving Tools** – Detecting and labeling human presence in stored media.
- **Low-Cost Surveillance Systems** – Strengthening existing camera networks with AI-driven detection.
- **Community or Research Projects** – Providing an open, adaptable tool for experiments in AI and HCI.

Limitations:

- **CPU-Bound Processing:** While a key advantage, the reliance on CPU can be slow for high-resolution images or images with many faces.
- **Static Image Only:** The current version only processes static images and does not support real-time video streams.
- **Single-Server Architecture:** The single AWS EC2 instance is not scalable for a large number of concurrent users and lacks high availability.
- **Limited Analysis:** The app only provides basic analysis (age, gender, emotion) and lacks deeper insights.

Future Enhancements:

- **Video Stream Support:** Extend the application to handle real-time video feeds for live surveillance or interactive use.
- **Automated Scaling:** Implement a more robust architecture using **AWS Auto Scaling Groups** and **Elastic Load Balancers** to handle increased user traffic.
- **Enhanced Analysis:** Integrate more advanced features such as facial landmark detection, emotion-based sentiment analysis, or face recognition.
- **Mobile and Edge-Friendly Versions:** Develop dedicated mobile applications (Android/iOS) or adapt the model for **IoT devices** for offline, edge-based processing.
- **Ethical Enhancements:** Add features like privacy modes and anonymization to address ethical concerns regarding facial recognition technology.

CONCLUSION:

The internship has been a highly enriching experience that provided me with both technical knowledge and practical exposure. Over the course of this program, I developed a deeper understanding of **cloud computing, web deployment, and machine learning concepts**, while also gaining hands-on experience with tools and frameworks used in real-world applications.

I explored critical aspects of **AWS Cloud** such as security, architecture, auto-scaling, and monitoring, which highlighted the importance of building reliable and cost-efficient systems on the cloud. In addition, I learned about the foundations of **data engineering**, including data pipelines, processing techniques, and how these elements integrate with machine learning to automate workflows.

Overall, the internship has provided me with the opportunity to merge **cloud computing, web development, and machine learning** into a complete project lifecycle — from development to deployment. Working on FaceIQ has enhanced my skills in designing scalable and secure AI applications, preparing me for future challenges in the fields of **cloud-based ML systems and intelligent web platforms**.

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